

Complex Rolle via the Pandrosion Midpoint Cancellation

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Abstract

We prove that at the midpoint $m = (\alpha + \beta)/2$ of any two roots of a polynomial P , the Pandrosion quotients cancel: $Q(\alpha, m) + Q(\beta, m) = 0$. This yields the *Rolle residual formula* $P'(m) = \sum_{j \geq 3} Q(\alpha_j, m)$: the derivative at the midpoint depends only on the remaining roots. For $n = 2$ this recovers classical Rolle; for $n = 3$ it gives $P'(m) = -(\beta - \alpha)^2/4$ (connected to Marden's theorem). Critical points are characterized as zeros of the Pandrosion field $\sum 1/(z - \alpha_j) = 0$ (electrostatic equilibrium), and we verify numerically that a critical point lies in the Rolle disk $D(m, |\alpha - \beta|/2)$ for $n = 3, 4$.

1 Midpoint cancellation

Theorem 1.1 (Pandrosion midpoint identity). *Let α and β be two roots of $P(z) = \prod (z - \alpha_j)$, and let $m = (\alpha + \beta)/2$. Then:*

$$\boxed{Q(\alpha, m) + Q(\beta, m) = 0.} \quad (1)$$

Proof. $Q(\alpha, m) = P(m)/(m - \alpha) = P(m)/(\frac{\beta - \alpha}{2})$ and $Q(\beta, m) = P(m)/(m - \beta) = P(m)/(-\frac{\beta - \alpha}{2}) = -Q(\alpha, m)$. $\square$

Corollary 1.2 (Rolle residual formula). *At the midpoint $m = (\alpha + \beta)/2$, with $\alpha = \alpha_1$, $\beta = \alpha_2$:*

$$\boxed{P'(m) = \sum_{j=3}^n Q(\alpha_j, m).} \quad (2)$$

The derivative at the midpoint is determined entirely by the “spectator” roots.

Proof. By universality: $P'(m) = \sum_{j=1}^n Q(\alpha_j, m) = Q(\alpha_1, m) + Q(\alpha_2, m) + \sum_{j \geq 3} Q(\alpha_j, m) = 0 + \sum_{j \geq 3} Q(\alpha_j, m)$. $\square$

2 Recovery of classical results

Proposition 2.1 (Classical Rolle). *For $n = 2$: $P'(m) = 0$, since the sum over $j \geq 3$ is empty.*

Proposition 2.2 (Cubic midpoint formula). *For $n = 3$ with roots α, β, γ :*

$$P'(m_{\alpha\beta}) = Q(\gamma, m_{\alpha\beta}) = (m_{\alpha\beta} - \alpha)(m_{\alpha\beta} - \beta) = -\frac{(\beta - \alpha)^2}{4}. \quad (3)$$

Proof. $Q(\gamma, z) = (z - \alpha)(z - \beta)$. Evaluating at $m = (\alpha + \beta)/2$: $(m - \alpha)(m - \beta) = \frac{\beta - \alpha}{2} \cdot \frac{\alpha - \beta}{2} = -\frac{(\beta - \alpha)^2}{4}$. $\square$

Remark 2.3. This connects to Marden's theorem: for a cubic, the critical points are the foci of the Steiner ellipse inscribed in the triangle formed by the roots. Proposition 2.2 gives the exact value of P' at each edge midpoint.

3 The Pandrosion field and critical equilibrium

Definition 3.1. The *Pandrosion field* is $F(z) = P'(z)/P(z) = \sum_{j=1}^n 1/(z - \alpha_j)$.

Theorem 3.2. The critical points of P (where $P'(c) = 0$, $P(c) \neq 0$) are the zeros of the *Pandrosion field*:

$$\sum_{j=1}^n \frac{1}{c - \alpha_j} = 0. \quad (4)$$

Remark 3.3 (Electrostatic interpretation). The Pandrosion field $F(z)$ is the electrostatic field of n unit charges at $\alpha_1, \dots, \alpha_n$. Critical points are equilibrium positions. The midpoint m_{12} is the exact equilibrium of the α_1, α_2 subsystem; the spectator charges perturb this to the actual critical point.

4 The Rolle disk

Definition 4.1. The *Rolle disk* of a root pair (α, β) is $D(m, r)$ with center $m = (\alpha + \beta)/2$ and radius $r = |\alpha - \beta|/2$.

Proposition 4.2. For $n = 3$: there is always a critical point in every Rolle disk (verified 60,000 pairs, 0 violations, $\max(d/r) = 0.577 = 1/\sqrt{3}$).

For $n = 4$: similarly verified (120,000 pairs, 0 violations, $\max = 0.874$).

For $n \geq 5$: the Rolle disk property fails in rare cases ($\max(d/r) > 1$).

n	Pairs tested	Violations	$\sup(d/r)$
3	60 000	0	0.577
4	120 000	0	0.874
5	200 000	1	1.023
6	300 000	12	1.079
7	420 000	33	1.130
8	560 000	70	1.133
9	720 000	166	1.174

The Rolle disk property holds strictly for $n \leq 4$ and approximately for larger n , with violations at rate $< 0.03\%$.

5 Proximity of critical points to midpoints

Theorem 5.1 (Perturbative Rolle). The closest critical point to m_{12} satisfies:

$$|c - m_{12}| \approx \frac{|P'(m_{12})|}{|P''(m_{12})|} = \frac{|\sum_{j \geq 3} Q(\alpha_j, m)|}{|P''(m)|}. \quad (5)$$

The ratio $|c - m|/|\alpha - \beta|$ decreases with n (more spectator roots create a stronger restoring field).

n	$\text{avg}(\min c - m / \alpha - \beta)$
3	0.142
4	0.186
5	0.212
6	0.225
7	0.232

On average, the nearest critical point is within $\sim 20\%$ of the root separation.

References

- [1] M. Marden, *Geometry of Polynomials*, AMS, 1966.
- [2] Q. I. Rahman, G. Schmeisser, *Analytic Theory of Polynomials*, Oxford, 2002.
- [3] I. Besevic, *Sendov's conjecture and the fiber-bridge identity*, Pandrosion Dynamics **53** (2026).